State space and observable algebras

Measurable state space

$$V = \mathbb{R}^d, \qquad \mathcal{B}(V) = \sigma(\text{Euclidean open sets}).$$

Linear maps on states

$$\mathrm{End}(V) \cong M_d(\mathbb{R}).$$

Addition and scalar multiplication are pointwise.

$$AB = A \circ B.$$

Associative and unital, with identity I_V.

Functions on states

$$\mathcal{A} = B_b(V).$$

Bounded Borel measurable real functions, with

$$(fg)(x) = f(x)g(x).$$

Commutative and unital, with identity $\mathbf{1}$.

Observable and probability evolution

Probability laws and the integration pairing

$$\mu \in \mathcal{P}(V), \qquad \langle f, \mu \rangle = \int_V f \, \mathrm{d}\mu = \mathbb{E}[f(X)], \quad X \sim \mu.$$

For a Borel measurable transformation $F : V \to V$, define

$$\mathsf{U}_F f = f \circ F,$$
$$\mathsf{T}_F \mu = F_{\#}\mu, \quad (F_{\#}\mu)(C) = \mu(F^{-1}(C)), \quad C \in \mathcal{B}(V).$$

$$\boxed{\langle \mathsf{U}_F f, \mu \rangle = \langle f, \mathsf{T}_F \mu \rangle.}$$

$\mathcal{P}(V)$ is convex. The pairing extends bilinearly to finite signed measures.

Project 9.1: the ODE and its solution spaces

Student Assessment Exercise, Project 9, item 1 (p. 12).

Let $A : I \to \mathrm{End}(V)$ and $B : I \to V$ be continuous. Solve

$$\dot{x}_t = A(t)x_t + B(t), \qquad x_s = x.$$

The ODE theorem gives a unique trajectory on I.

A linear operator on trajectories

$$\mathcal{X} = C^1(I, V), \qquad \mathcal{Y} = C(I, V),$$
$$\mathcal{L} : \mathcal{X} \to \mathcal{Y}, \qquad (\mathcal{L}\gamma)(t) = \dot{\gamma}(t) - A(t)\gamma(t).$$

The homogeneous solutions form the vector space

$$\mathcal{H} = \ker \mathcal{L} = \{\eta \in \mathcal{X} : \dot{\eta} = A\eta\}.$$

Homogeneous evolution

Evaluation at time s identifies trajectories with states:

$$E_s : \mathcal{H} \to V, \quad \eta \mapsto \eta(s), \qquad \mathcal{H} \cong V, \quad \dim \mathcal{H} = d.$$

Existence makes E_s onto. Uniqueness makes it one-to-one.

Evolution on states

$$\begin{aligned} U(t,s) &= E_t E_s^{-1}, & U(s,s) &= I_V, \\ \partial_t U(t,s) &= A(t)\, U(t,s), & & \\ U(t,r)\, U(r,s) &= U(t,s), & U(t,s)^{-1} &= U(s,t). \end{aligned}$$

Variation of constants and verification

Seek a particular solution $\gamma_p(t) = U(t,s)c(t)$ with $c(s) = 0$.

$$\dot{\gamma}_p = A(t)\gamma_p + U(t,s)\dot{c},$$
$$U(t,s)\dot{c}(t) = B(t) \implies \dot{c}(t) = U(s,t)B(t).$$

Integrate and use composition to obtain

$$\boxed{x_t = U(t,s)x + \int_s^t U(t,r)B(r)\,\mathrm{d}r.}$$

Verification $U(s,s) = I_V$ gives $x_s = x$. The Leibniz rule gives

$$\dot{x}_t = A(t)\left[U(t,s)x + \int_s^t U(t,r)B(r)\,\mathrm{d}r\right] + B(t) = A(t)x_t + B(t).$$